Computer Networks - Lecture 1
Instructor: Prof. Claude Shannon

1. The OSI 7-Layer Reference Model
The OSI model partitions a telecommunication system into seven abstraction layers:
- Layer 7: Application (HTTP, DNS, SSH)
- Layer 4: Transport (TCP reliable byte stream, UDP connectionless datagrams)
- Layer 3: Network (IP logical addressing and packet forwarding)
- Layer 2: Data Link (Ethernet frames, MAC addressing)
- Layer 1: Physical (Bits, transmission media, electromagnetic signals)

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2. Routing Protocols & Algorithms
Routing is the process of selecting paths across multiple networks.
Two dominant routing paradigms:
- Link-State Routing: Every router computes Dijkstra's algorithm to find shortest paths (e.g., OSPF).
- Distance-Vector Routing: Routers exchange distance estimates using the Bellman-Ford algorithm (e.g., RIP). C